Collective Seed Storage

A Sustainable Approach to Ex Situ Seed Conservation through Citizen Participation and Distributed Storage

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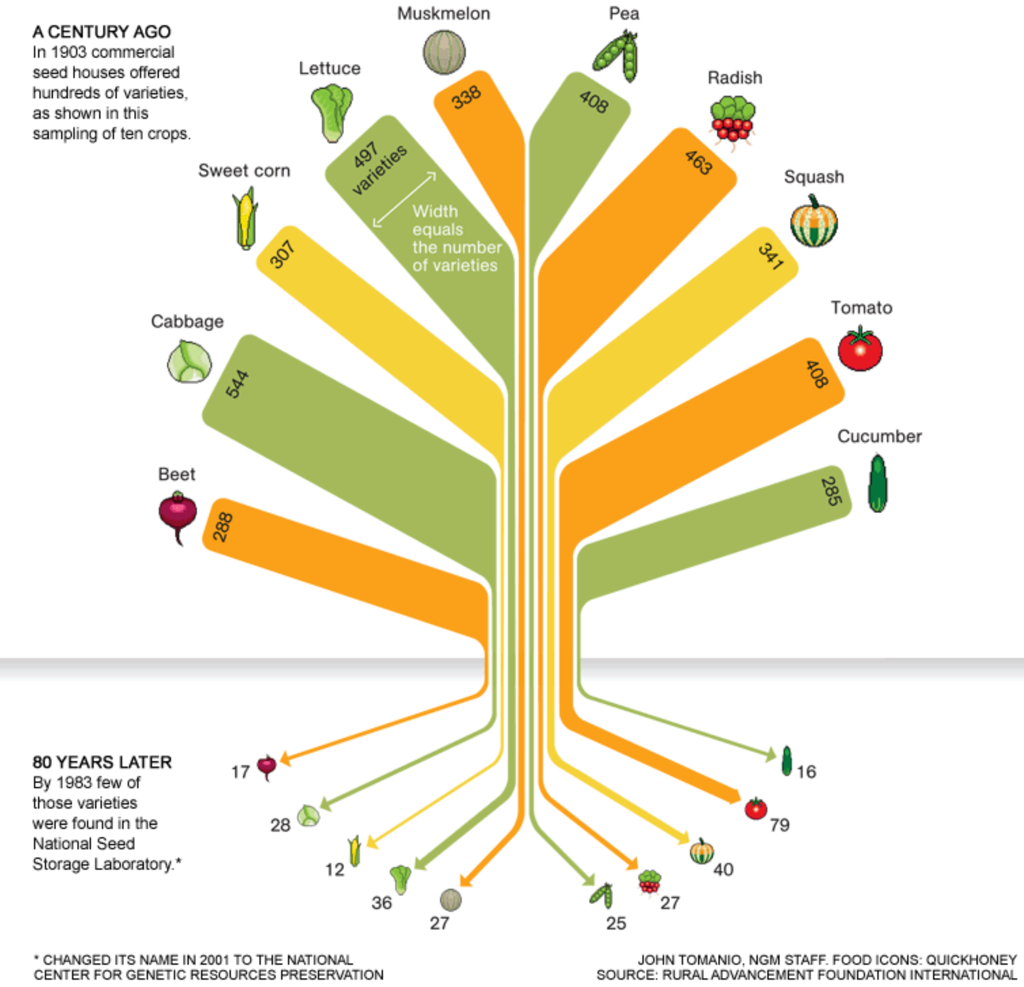
The Need for Seed Diversity

- Over 50,000 edible plants exist worldwide, but **just 15 provide 90% of food energy intake** (e.g., rice, corn, wheat).
- More than 95% of crop **genetic erosion** articles report changes in diversity, with nearly 80% showing evidence of loss [1].

🌱 Genetic diversity

Vital to adapt to climate change, extreme weather, pests, and diseases



Conservation Strategies

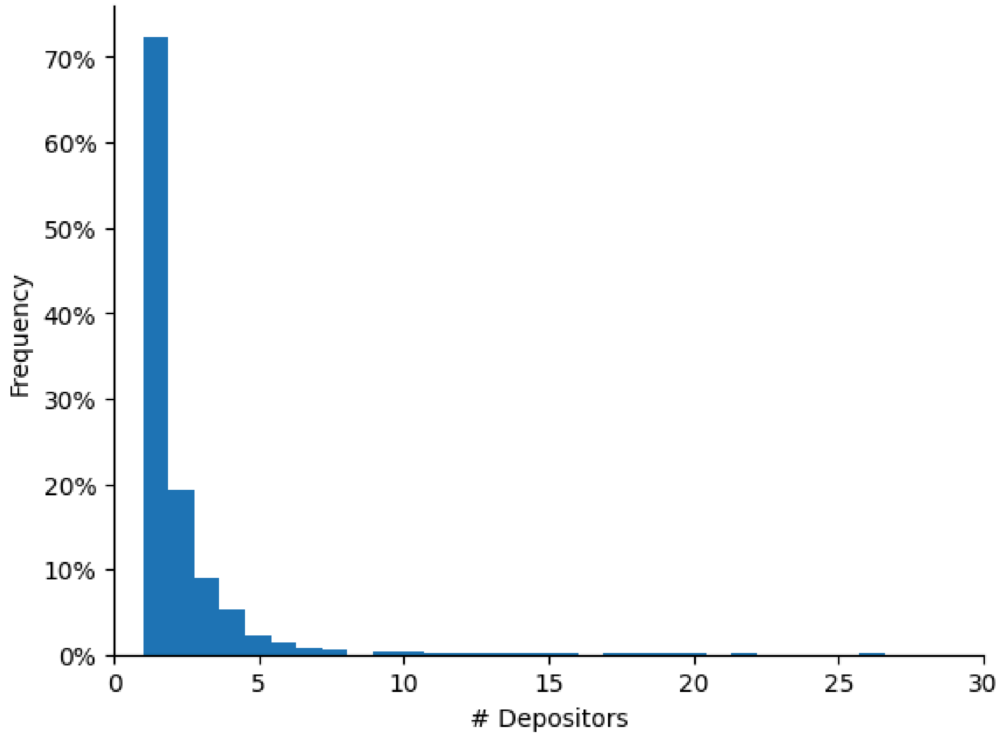
Criteria	In Situ Conservation	Ex Situ Conservation
Location	Natural environment / farms	Seed banks, seed vaults
Evolution	Plants adapt naturally	Static; no natural evolution
Risk of Loss	Higher (climate, land-use)	Lower (controlled conditions)
Access	Local and informal	Global access via formal mechanisms
Cost	Lower short-term cost	Higher cost (infrastructure)
Best Use	Wild relatives, landraces	Major crops, backup

Svalbard Global Seed Vault (SGSV)

- A global backup facility for seed banks worldwide.
- Safeguards over 1.2 million samples representing >6,000 plant species.
- Unanimously considered the **backup seed facility that provides the highest standards of safety and security.**

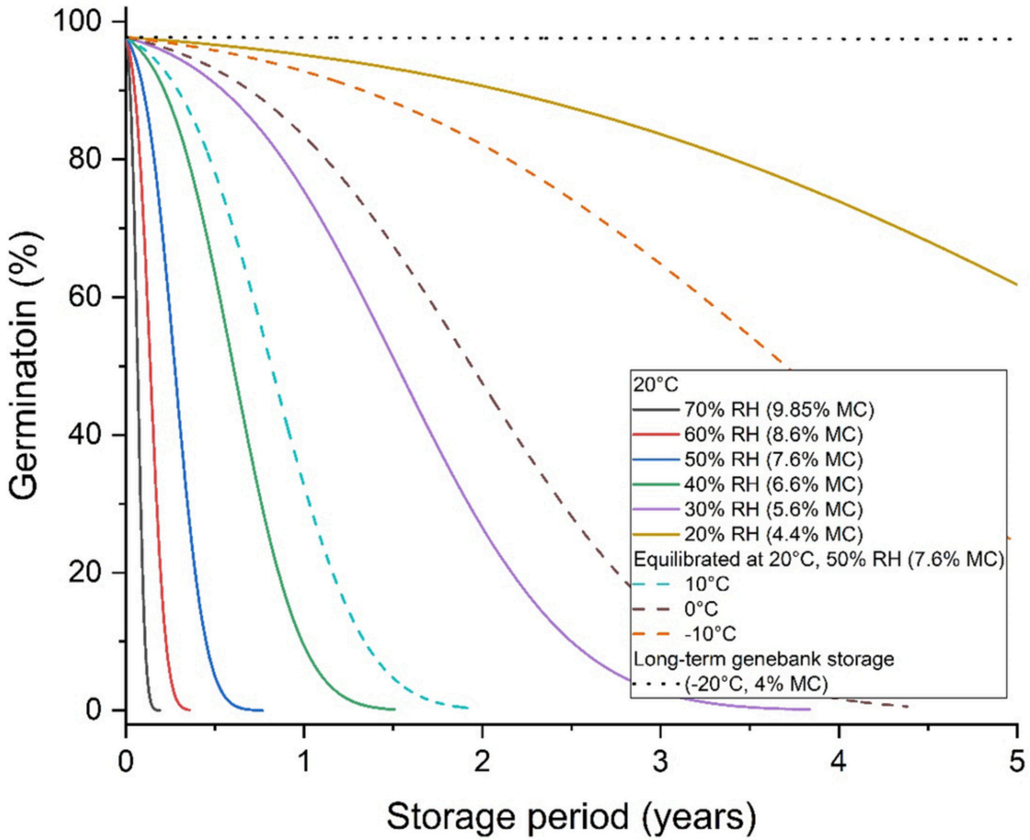
Open Problem

- The vast majority of species, mostly CWR in the SGSV, have only a **single depositor** [2].
- **The Threat:** if a disaster hits both the primary seed bank and the vault, the species could be lost.



Orthodox Seeds and Storage

- **Orthodox Seeds:** Approximately 90% of species can retain viability if dried and kept at low temperatures (e.g., wheat, rice, legumes).
- **Storage Standards:** FAO recommends [3] long-term storage at **-18°C and 15% relative humidity.**
- Subzero freezers are acceptable if ideal conditions are not available.



Collective Seed Storage (CSS)

- **Domestic freezers provide suitable conditions for seed conservation**
- **Vision:** Each household becomes a micro-node in a distributed seed conservation system. Citizens are active contributors to food system resilience and biodiversity conservation
- **Shared Economy Approach:** Sharing a tiny, idle portion of home freezers for seed preservation, to drastically increase redundancy and the overall availability of seeds with minimal infrastructure investments.

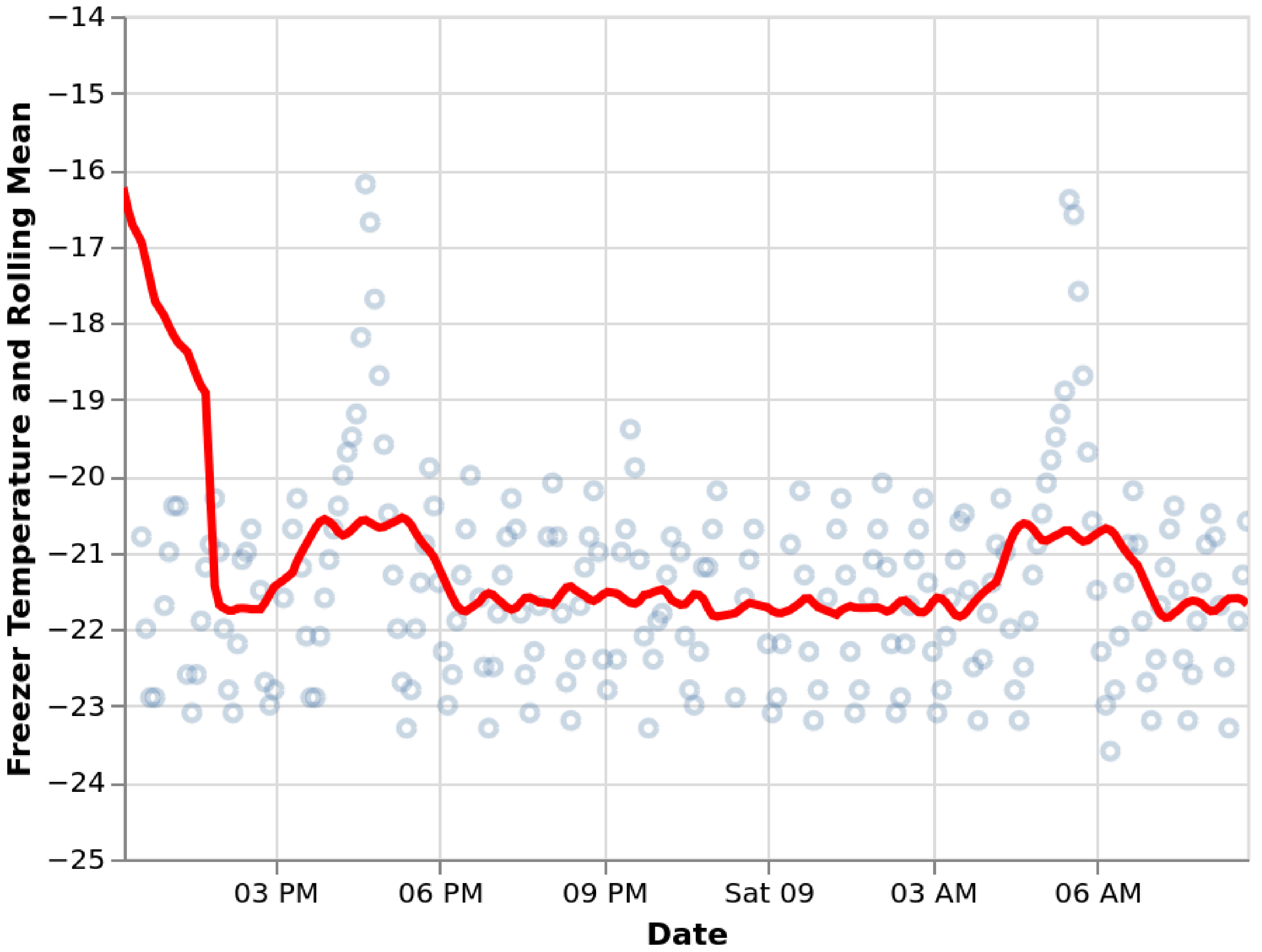
CSS Requirements

To ensure feasibility and active participation, the system must meet:

1. **Storage Standards (R1):** FAO long-term storage conditions (approx. -18°C). Subzero acceptable.
2. **User-Friendly (R2):** Simple installation using off-the-shelf equipment and home WiFi.
3. **Longevity (R3):** Operate for at least 1 year without user intervention (battery life).
4. **Incentivization (R4):** Implement mechanisms to reward users for their long-term commitment.

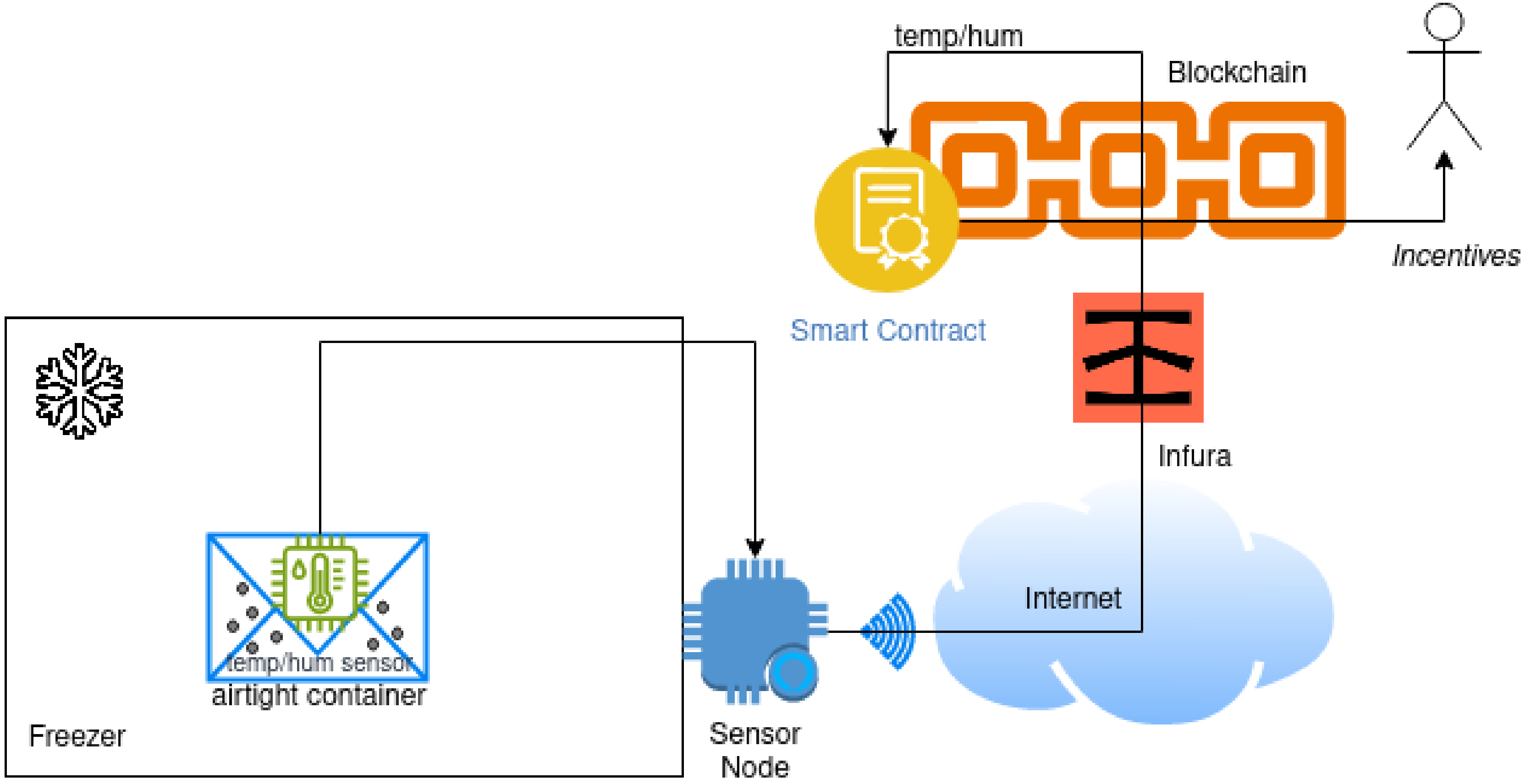
R1: Storage

- FAO standards 😐
- Subzero conditions 🙂



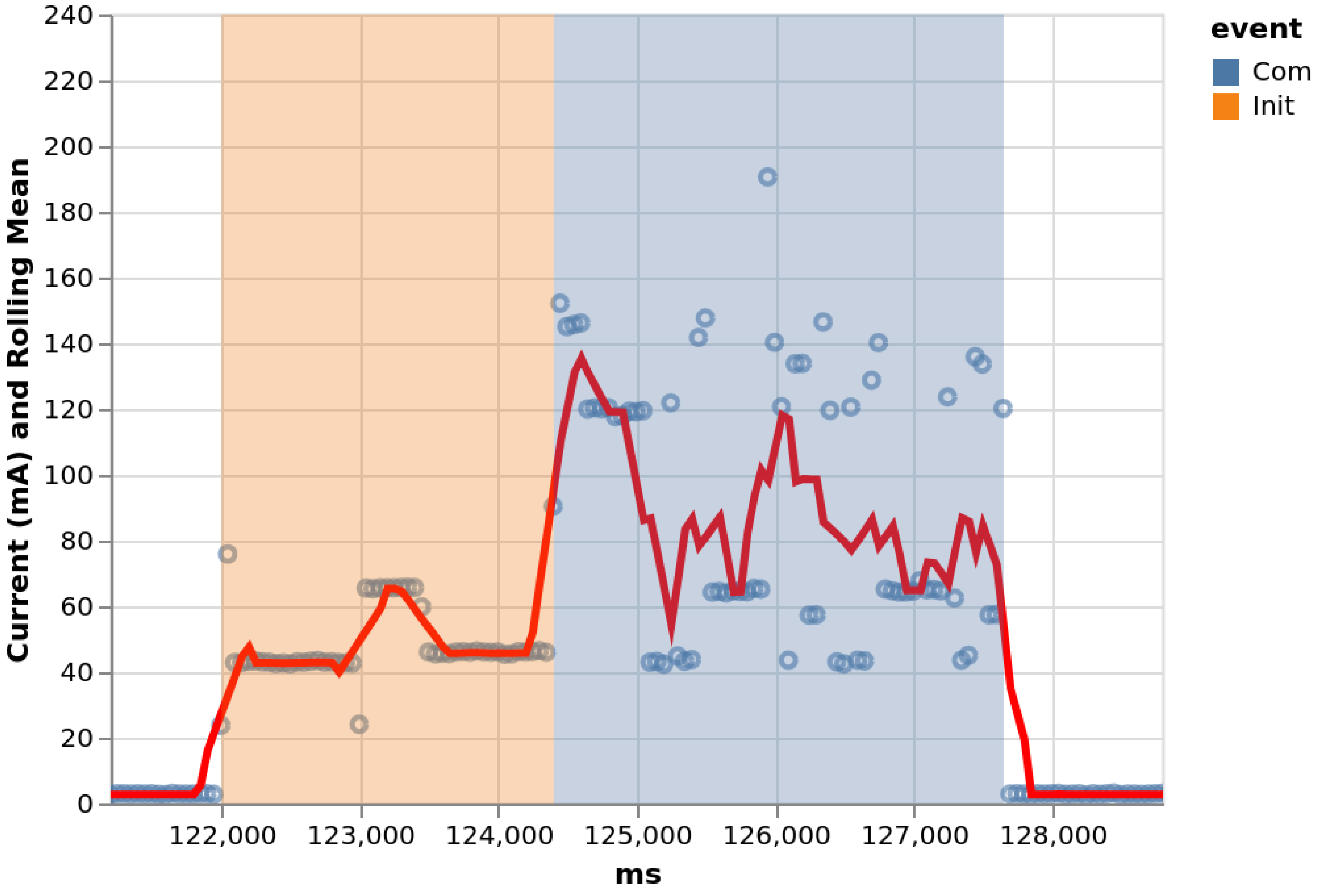
Architecture

- R2: friendly 🙂



R3: Lifetime

- **> 1 year** 🙂
- 1.5Ah LiFePo4 battery
- Duty Cycle: active period about 6 seconds, sleep period >0.73 hours



R4: Incentive

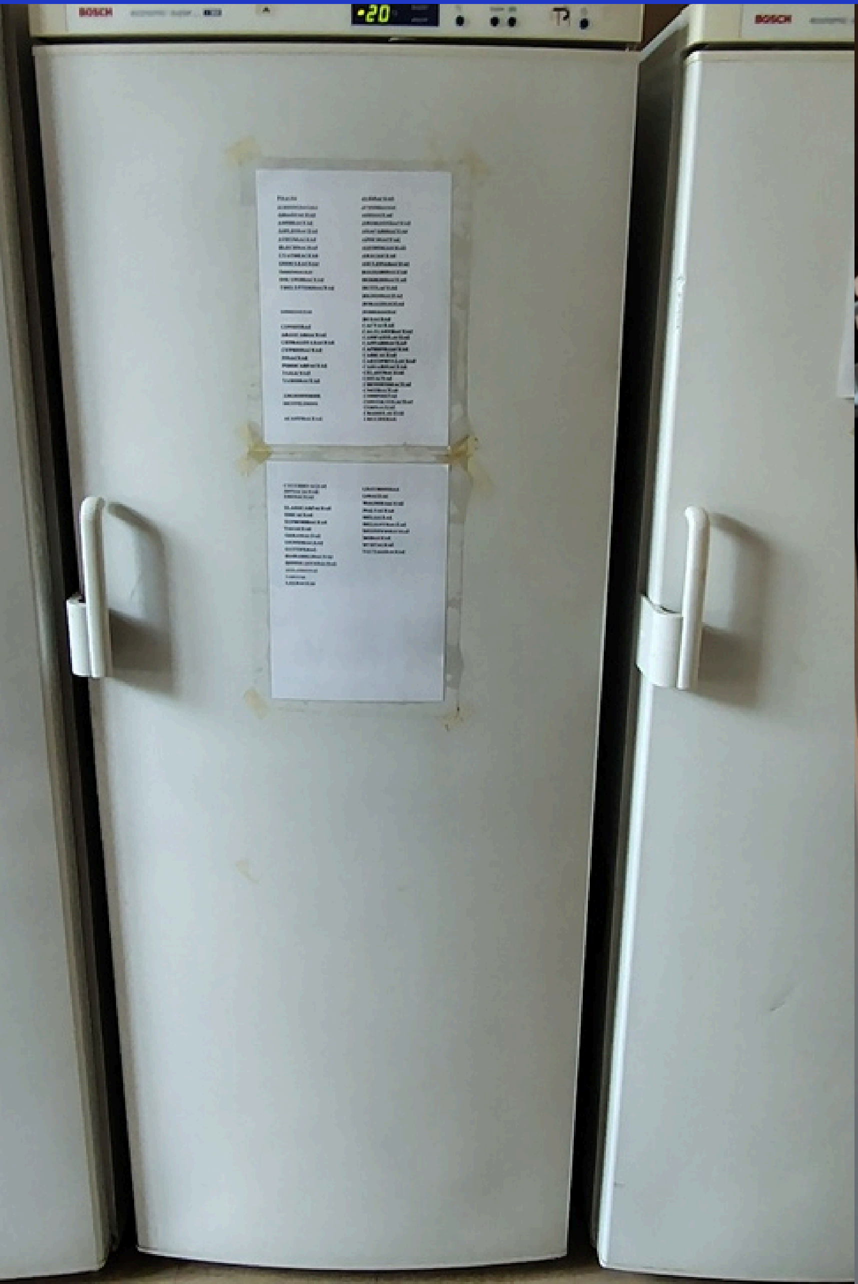
- Blockchain Lottery 😐

Conclusions

- CSS can support the preservation of seed diversity by transforming urban citizens into "active" agents of agricultural resilience.
- Households become micro-nodes within a distributed seed conservation system
- Our Proof-of-Concept demonstrates the feasibility of the core technical components underlying this vision.


SUSTAINABLE DEVELOPMENT GOALS
1 NO POVERTY
2 ZERO HUNGER
3 GOOD HEALTH AND WELL-BEING
4 QUALITY EDUCATION
5 GENDER EQUALITY
6 CLEAN WATER AND SANITATION
7 AFFORDABLE AND CLEAN ENERGY
8 DECENT WORK AND ECONOMIC GROWTH
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
10 REDUCED INEQUALITIES
11 SUSTAINABLE CITIES AND COMMUNITIES
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
13 CLIMATE ACTION
14 LIFE BELOW WATER
15 LIFE ON LAND
16 PEACE, JUSTICE AND STRONG INSTITUTIONS
17 PARTNERSHIPS FOR THE GOALS

A recent visit to the seed bank at Rome's Botanic Garden



Discussion

- A modern approach to seed preservation should promote more active citizen participation (e.g., Seed Guardians) that extends beyond the mere act of storage.
- Although the preservation of seeds in domestic freezers is a novel and valuable practice, it may remain, to some extent, a "passive" activity.

Future Vision: www.seedpeers.net

Thank you!

Questions?

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https://andreavitaletti.github.io/

https://github.com/andreavitaletti/slides/tree/main/ICAISF
2026

Image sources

- Slide 6: https://doi.org/10.1111/rec.13174
- Slide 2: https://medium.com/thenextnorm/importance-of-genetic-diversity-in-agriculture-b9f88f5fda55
- Slide 4: https://en.wikipedia.org/wiki/Svalbard_Global_Seed_Vault#/media/File:Svalbard_Global_Seed_Vault_February_2025.jpg